D03 · Co-Manufacturing Operations

05 · Multi-site coordination & capacity allocation

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1 WHAT IT IS

Multi-site coordination and capacity allocation across co-manufacturers is the discipline of designing the production-volume distribution across a portfolio of contract manufacturers as a risk-economics problem rather than a unit-cost problem — incorporating not only landed unit cost but also reliability differential (track record on quality, delivery, capability), concentration risk (exposure to single-CMO failure), cost-to-switch (qualification cost, lead time, capability loss), and operational complexity (coordination overhead, IT integration, multi-site governance). The economic objective is minimum total expected cost across the portfolio under disruption scenarios, not minimum unit cost in steady state — and these two formulations produce fundamentally different capacity allocations.

2 WHY IT MATTERS

Multi-CMO capacity allocation is the strategic decision in any co-manufacturing portfolio that produces the largest difference between operational resilience and operational fragility — and the decision most consistently made on the wrong criteria. **The dominant failure mode is capacity allocated to CMOs based on unit-cost competitiveness, with explicit "diversification" claimed because volume runs across multiple CMOs, but the underlying allocation pattern concentrates 60–80% of volume on 1–2 lowest-cost CMOs. When one of those CMOs experiences a disruption — regulatory finding, capacity loss, quality crisis, ownership change — the operation discovers that the paper diversification was concentration in practice, and the qualification cost and lead time of moving volume to alternates measure in quarters not weeks**. The peer-reviewed academic work on supply chain risk management is explicit on this structural error **(Sodhi & Tang, 2012 — peer-reviewed, "Managing Supply Chain Risk", Springer)**: capacity allocation across multiple suppliers must explicitly model concentration risk and switching economics, not just unit cost, and operations that optimize on cost alone systematically accumulate disruption exposure that materializes 1–3x per decade per major portfolio. Industry data confirms — **operations with multi-criterion allocation (cost + reliability + concentration + switch cost) absorb major CMO disruptions with 2–8 weeks of operational impact at 3–8% incremental cost; operations with cost-only allocation absorb the same disruptions with 12–36 weeks of operational impact at 25–60% incremental cost during the disruption window, plus catastrophic revenue exposure**. The cost saved in steady state is paid back at high interest during disruption.

3 HOW IT IS DONE

1. **Frame allocation as a multi-criterion optimization — not as a cost-minimization with diversification constraints.** The optimization objective is expected total cost across the portfolio over a 3–5 year horizon, including disruption scenarios with documented probabilities. The criteria entering the optimization: landed unit cost, demonstrated reliability differential (quality, delivery, capability), concentration risk (exposure if this CMO fails), cost-to-switch (qualification cost, lead time, capability loss if migration is needed), and operational complexity overhead (coordination, governance, IT integration). Plants that frame allocation as cost-minimization-with-diversification-as-constraint produce concentration patterns; plants that frame it as multi-criterion produce resilient portfolios at slightly higher steady-state cost.
2. **Set explicit concentration limits — not implicit ones that get violated under cost pressure.** Concentration limits should be expressed as hard constraints in the allocation framework: no single CMO greater than X% of total revenue (typically 30–40% for non-critical categories, 20–30% for critical), no single SKU 100% concentrated at one CMO (every commercial SKU should have at least one qualified alternate), no critical product family without backup capacity at a second qualified site. Implicit diversification preferences fail under cost pressure because the cost case for shifting marginal volume to the lowest-cost CMO is always present; explicit hard constraints survive cost pressure because they require explicit override at the leadership level.
3. **Quantify cost-to-switch explicitly — qualification cost and lead time are real assets that the portfolio either has or does not have.** A CMO that could be qualified for additional product families in 6 months at $200K is a different asset than a CMO that would require 14 months and $800K. The cost-to-switch differential should be modeled in the allocation framework as the cost of building optionality, and the portfolio should maintain pre-qualified backup capacity on critical product families even when current volume does not justify it. The pre-qualification cost is portfolio insurance; operations that decline to pay it

during steady state pay much more during disruption.

4. **Model disruption scenarios explicitly — not as hypothetical possibilities but as expected events with documented frequencies.** Major CMO disruptions (regulatory action, capacity loss, quality crisis, ownership change, force majeure) occur at industry-documented frequencies that can be estimated from regulatory databases, industry data, and historical experience. A CMO portfolio with N partners over a 5-year horizon will typically experience 0.5–2 major disruption events; the allocation framework should explicitly compute expected cost under those scenarios and select the allocation that minimizes expected total cost rather than steady-state cost. Plants that treat disruption as not-going-to-happen produce allocations that price disruption at zero; plants that price disruption at its actual frequency produce different and more resilient allocations.
5. **Operate the portfolio with active rebalancing capability — not as a static allocation set at qualification time.** Multi-CMO portfolios that are designed for resilience but never exercised lose the capability to rebalance under disruption. The portfolio should be operated with periodic deliberate rebalancing (semi-annual or annual) that shifts volume across CMOs based on demonstrated reliability, capability development, and capacity availability. This exercises the rebalancing infrastructure (qualification status, IT integration, contractual terms, operational coordination) so that when disruption requires emergency rebalancing, the muscle is real. Plants that design resilient portfolios but never rebalance discover at disruption time that the optionality was theoretical.
6. **Govern at the operations + supply chain + finance + risk level — not at procurement alone.** Capacity allocation decisions have implications across functions: operations (production reliability, quality consistency), supply chain (capacity planning, inventory positioning), finance (working capital, disruption-cost exposure), risk (concentration risk, regulatory exposure). Procurement-only governance produces cost-optimal allocations because procurement's metrics reward cost reduction. Multi-functional governance produces risk-economics-optimal allocations because the additional functions value the criteria that cost-only optimization ignores. Plants that establish multi-functional governance produce resilient portfolios; plants that defer to procurement-led allocation produce concentration that performs well in steady state and badly under stress.

WORKED EXAMPLE — ILLUSTRATIVE

A nutraceutical brand sells 8 SKUs across 4 contract manufacturers; the allocation evolved over time based on landed unit cost negotiations with each CMO. **Current allocation: 73% of total volume runs through 2 CMOs (CMO-A at 41%, CMO-B at 32%); 4 SKUs representing 64% of revenue are 100% concentrated at single CMOs; "diversification" is claimed because 4 CMOs are in the portfolio**. CMO-A receives an FDA Form 483 with significant compliance findings; regulatory remediation is expected to take 6–9 months with restricted production capacity during remediation. The operation has no qualified backup with available capacity for the 3 SKUs concentrated 100% at CMO-A; production stops on SKUs representing 28% of revenue immediately and another 12% within 6 weeks as buffer inventory depletes; the brand spends 4 months running emergency qualifications at CMO-C and an entirely new CMO-E at expedited cost, with $14M in lost revenue and $3.2M in expedited qualification costs. The restructured allocation framework: **(1) multi-criterion optimization with explicit modeling of unit cost, reliability differential, concentration risk, cost-to-switch, and disruption-scenario expected cost over 3 years; (2) hard concentration limits — no CMO greater than 30% of revenue, no SKU 100% concentrated at a single CMO, pre-qualified backup capacity for every critical SKU; (3) cost-to-switch quantification and active investment in pre-qualification of backup capacity at $480K total cost; (4) disruption-scenario modeling using industry-documented CMO event frequencies; (5) annual deliberate rebalancing exercising the multi-CMO infrastructure; (6) multi-functional governance through operations + supply chain + finance + risk leadership**. The economic comparison: **landed unit cost increases 3.2% across the portfolio under the new allocation (~$1.4M/year on the volume base); pre-qualification investment $480K one-time; total annual carry of resilience ~$1.9M**. Two years after restructure, a different CMO in the portfolio experiences a compliance issue requiring 30% of its volume to be reallocated; under the restructured allocation framework, the volume reallocates in 6 weeks with no production stops and no expedited qualification cost — the resilience that cost $3.8M to carry for two years saves $14M+ in a single disruption event.

IN PRACTICE

Frame allocation as multi-criterion optimization, not cost-with-constraints

Cost + reliability + concentration + switch cost as joint dimensions, not cost-minimization with diversification rules bolted on **(Sodhi & Tang, 2012 — peer-reviewed, Springer)**. The bolt-on approach systematically loses the optimization battle to procurement metrics every quarter.

Set explicit hard concentration limits

No CMO greater than 30% of revenue; no SKU 100% concentrated at a single CMO. These are structural constraints that survive cost-pressure reviews because they were committed before the cost-pressure review opened. Soft guidelines that "recommend" diversification get traded away.

Periodic deliberate rebalancing to exercise the infrastructure

The pre-qualified backup capacity that has never produced production volume is theoretical capacity. Deliberate rebalancing — moving 5–15% of volume periodically to backup CMOs — keeps the qualification current, the relationship live, and the actual switch cost honest. The infrastructure that is never used does not work when needed.

Joint governance: operations + supply chain + finance + risk

Procurement and finance metrics reward concentration; without explicit governance giving operations and risk veto on concentration-violating allocations, procurement-finance optimization wins every decision until a disruption makes the cost of those wins visible.

WHAT YOU WOULD USE

MULTI-SITE PORTFOLIO COORDINATION

Coupa SCD (former LLamasoft) / OMP / Logility — supply chain network design and optimization — Network-design platforms with multi-site capacity-allocation optimization, disruption-scenario modeling, and total-cost-of-ownership analysis across supplier portfolios; the right fit for operations with 10+ CMOs and complex multi-SKU allocation decisions **(Coupa — vendor; OMP — vendor; Logility — vendor)**.

Interos / Resilinc / Everstream Analytics — supply chain risk and resilience platforms — Specialized risk platforms that monitor supplier risk signals (regulatory, financial, geopolitical, climate, reputational) and quantify concentration risk and disruption exposure across CMO portfolios; the right fit for operations adding risk-intelligence layers to existing procurement infrastructure **(Interos — vendor; Resilinc — vendor; Everstream — vendor)**.

SAP IBP / Kinaxis RapidResponse / o9 Solutions — integrated business planning with allocation — Enterprise IBP platforms that integrate demand planning, capacity planning, and supplier allocation, supporting dynamic rebalancing across CMOs as demand and capacity signals shift; the right fit for operations whose CMO allocation must be integrated with broader S&OP processes **(SAP — vendor; Kinaxis — vendor; o9 Solutions — vendor)**.

A documented multi-criterion allocation framework + explicit concentration limits + periodic deliberate rebalancing + multi-functional governance — For operations without enterprise platform investment — an Excel-based allocation framework with explicit modeling of cost + reliability + concentration + switch cost + disruption scenarios, hard concentration limits documented in governance, annual or semi-annual rebalancing exercise, joint governance across operations + supply chain + finance + risk. The lowest-tech option produces strong outcomes when the framework rigor is real and the organizational backing is consistent.

4 SO WHAT

Allocating volume to co-manufacturers by lowest unit cost is saving 2–4% in good times and losing 30–70% of revenue when a CMO falls — the paper diversification becomes real concentration under cost pressure. Frame allocation as multi-criterion optimization not cost-minimization with diversification constraints, set explicit hard concentration limits (no CMO greater than 30% revenue, no SKU 100% concentrated), quantify cost-to-switch and invest actively in pre-qualified backup capacity, model disruption scenarios at documented industry frequencies, operate the portfolio with periodic deliberate rebalancing to exercise the infrastructure, and govern across operations + supply chain + finance + risk not procurement alone. The 2–5% resilience cost carried in steady state buys disruption protection worth 10–30x its annual cost when disruption occurs — and disruption occurs 1–3 times per decade in any major CMO portfolio.

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